Posture Guard – A Human Sitting Posture Detection And Alarming System

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Aman Jain
Department of CSE & IT, Jaypee Institute of Information Technology, Noida, India,
amanjn2003@gmail.com

Shikha Jain
Department of CSE & IT, Jaypee Institute of Information Technology, Noida, India,
shi_81@rediffmail.com

Abstract* - Poor posture due to prolonged screen time is an increasingly prevalent health issue with long-term consequences such as musculoskeletal discomfort and pain. To address this issue, we introduce the "Posture Guard" system, an innovative solution that leverages computer vision, machine learning, and real-time feedback to monitor and improve posture. Utilizing a webcam, the system analyzes users' posture in real-time, providing immediate visual and audio feedback for corrective action. First, data is collected under five classes: namely, "Normal", "Neck", "Shoulders", "Forward Bend", "Backward Bend". Then, multiclass CNN is trained to classify the data into different classes. This paper discusses the development and evaluation of the Posture Guard system, highlighting its key features which include robust posture analysis, data tracking, and user-friendly interface. This study employed a three-phase testing approach to evaluate the robustness of this system in various scenarios like different environment and lighting conditions. The model derived through this research gave around 99% precision on people who were a part of training data and gave an average precision of 64% on unseen faces in a different environment. The system demonstrated the potential to substantially improve posture habits, as evidenced by trials showing enhanced awareness and significant posture correction among users. The implications of this study suggest that such technological interventions could mitigate health risks and improve the quality of life for users engaged in sedentary activities.

Keyword- Posture detection, Teleworking, ergonomics, Musculoskeletal health, Real-time video processing, Human pose estimation, Automated posture assessment, Health monitoring

I. Introduction

Prolonged periods of sedentary behavior, particularly common in contemporary professional and academic environments, are a substantial risk factor for the development of poor posture. Consequences of such posture include a spectrum of musculoskeletal conditions with potential long-term health implications. Furthermore, the increasing screen time in post-pandemic society has accentuated this issue, creating a demand for effective preventive strategies [1]. Despite growing public awareness of the health risks associated with sedentary lifestyles, practical solutions to encourage proper posture are limited.

The typical approach to mitigating this issue has primarily focused on ergonomic furniture and educational programs, which have not sufficiently addressed the need for continuous, personalized posture monitoring.

The state-of-art in the field indicates that while there has been progress in the development of technologies for posture monitoring, challenges remain. These include achieving a balance between accuracy, user compliance, and real-time feedback. Moreover, no dataset is available publicly for front sitting posture to help the people sitting in front of the computer. The research presented in this paper intends to fill this gap by delivering a non-invasive, real-time, and interactive posture monitoring and correction system that aligns closely with the user's lifestyle and technology usage patterns.

Recent advancements in computer vision and machine learning technologies offer promising avenues for real-time posture detection and correction. Such technologies can provide immediate, non-invasive feedback, encouraging users to self-correct their posture more effectively than previous interventions. This research seeks to motivate the use of Convolutional Neural Networks (CNN's) for classifying postures into "Good" and "Bad" classes or posture specific classes as discussed in this paper (shown in Figure 1). CNN's have been out casting all other models when it comes to task related to images, which makes it an ideal choice for our problem statement. Since, like other researches on this problem statement, we are not using any external hardware and simply relying on the webcam data, CNN model will be the backbone of our whole system.

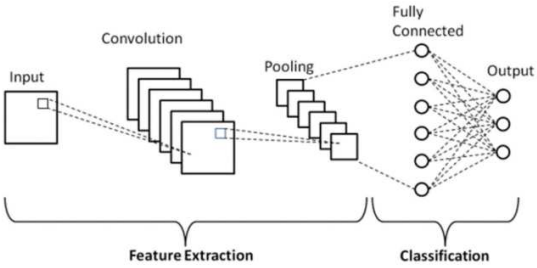

Fig 1: Deep Convolutional Neural Network Architecture [2]

This paper presents "Posture Guard," a cutting-edge system designed to detect poor posture and provide instant feedback for correction, thereby aiming to reduce the prevalence of musculoskeletal complications and improve overall health. We critically assess existing posture correction methodologies, highlighting their inadequacies such as delayed feedback and limited user engagement.

Herein, we lay the foundation for a novel, technology-driven solution that juxtaposes real-time monitoring with a user-friendly interactive interface. Posture Guard integrates sophisticated computer vision algorithms with ergonomic principles to facilitate sustained postural health, aligning technical innovation with ease of use to encourage widespread uptake.

In the proceeding sections, we elaborate on the methodology, deployment, and efficacy of Posture Guard,

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with the ultimate aim of demonstrating the significant role technology can play in ameliorating the pervasive issue of poor posture. By examining the reduction in posture-related health problems among users of the system, we aim to highlight both the individual and collective benefits of this intervention.

II. Literature Review

Poor posture is identified as a major contributor to musculoskeletal disorders, which are among the most common causes of severe long-term pain and physical disability. Similar to the present study, few studies emphasized the financial implications of MSDs resulting from poor ergonomics, drawing attention to the high costs for healthcare systems globally. Furthermore, Patel and Kumar detailed the psychosocial impacts, including decreased work productivity and quality of life [3]. Ergonomic interventions, such as adjustable chairs and desks, have been widespread but show mixed results in terms of long-term posture improvement. In response, researchers have explored wearable technologies for posture monitoring as seen in the studies [4, 5]. While promising, these solutions often suffer from user compliance issues and discomfort over extended wear. The application of computer vision in health monitoring has surged in recent years, with Zhao et al. demonstrating its viability in detecting real-time human activity. Building on this foundation, Gomez et al. specifically applied computer vision for posture tracking, reporting significant advancement but also acknowledging the need for greater accuracy and user engagement [6, 7].

One study delves into the impact of teleworking on workers' posture and proposes an automated system to detect and quantify incorrect postural habits, offering preventive recommendations to mitigate associated health risks. With the surge in telecommuting due to the COVID-19 pandemic, concerns have risen regarding its potential effects on musculoskeletal health. Employing a specialized hardware system integrated with convolutional neural networks (CNNs), the authors developed a real-time posture detection system. Through real-time video processing, the system detects postural deviations in workers' necks, shoulders, and arms, issuing timely recommendations to avert potential health complications. Key findings underscore the prevalence of incorrect postures among teleworkers, posing an elevated risk of musculoskeletal disorders. The study emphasizes the significance of maintaining proper postural hygiene to prevent spinal injuries and other related conditions. Evaluation of the system's performance across diverse hardware platforms demonstrates its efficacy in real-world scenarios, thereby addressing the pressing need for automated tools to safeguard workers' postural health in the era of remote work [8].

One study introduces a novel deep recurrent hierarchical network (DRHN) model based on MobileNetV2 for human posture detection, with a specific focus on recognizing sitting postures. Emphasizing the critical role of posture detection across various domains such as assisted living, healthcare, and physical exercise, the authors identify shortcomings in existing methodologies, particularly in addressing occlusion challenges when the full human skeleton is not visible in the frame. The proposed DRHN architecture processes RGB-Depth frame sequences to generate semantically related posture representations. By analyzing temporal features from video frames and employing hierarchical data representation, the model effectively mitigates occlusion issues associated with the limited visibility of the human torso. A pilot study involving 11 participants validates the model's efficacy, achieving an impressive 91.47% accuracy in sitting posture recognition at a rate of 10 frames per second. The study underscores the significance of posture recognition in mitigating medical conditions linked to sedentary lifestyles, such as back pain and work-related musculoskeletal disorders. Moreover, the authors curate a dataset comprising 66 unique video sequences, facilitating model training and evaluation. Results demonstrate the proposed approach's viability for real-time applications. Comparative analysis with existing posture recognition methods highlights the lightweight nature of the DRHN model, rendering it suitable for real-time deployment. Addressing limitations such as sample size constraints and the need for diverse subject groups, the study advocates for ongoing enhancements in dataset collection methodologies. Furthermore, insights into computational aspects, including training optimization and data augmentation, enrich the discussion on methodological advancements. In conclusion, the DRHN model presents a promising avenue for real-time sitting posture recognition, effectively addressing occlusion challenges and achieving high classification accuracy. The study's meticulous methodology and validation procedures contribute to advancing human posture detection, with potential applications in assisted living and healthcare domains [9].

One paper meticulously addresses the intricate challenges associated with employing deep Convolutional Neural Network (CNN) models for human posture recognition. Presenting a novel three-phase model, it aims to provide decision support to effectively mitigate these hurdles. Central to the study is the utilization of transfer learning, image data augmentation, and hyperparameter optimization (HPO) techniques to enhance classification performance across renowned models like AlexNet, VGG16, CNN, and MLP, leveraging the MPII human pose dataset. Key achievements include remarkable accuracy rates, such as 91.2% with AlexNet, 90.2% with VGG16, 87.5% with CNN, and 89.9% with MLP. These results underscore the efficacy of the proposed methodologies in addressing the inherent complexities of deep CNN models. Notably, the study marks a significant milestone as the first to implement HPO on the MPII human pose dataset, thereby breaking new ground in posture recognition research. The paper meticulously navigates the challenges posed by deep CNN models, ranging from overfitting to suboptimal performance, particularly evident with expansive models like AlexNet and VGG16. To overcome these challenges, the study advocates for integrating CNN transfer learning, image data augmentation, and HPO into a cohesive three-phase model. It highlights the transformative potential of HPO methods in shaping training algorithm behavior and influencing model performance metrics. Moreover, the paper presents implementation results of transfer learning models, both independently and in conjunction with image augmentation or HPO. Through rigorous experimentation, it elucidates the nuanced impact of different hyperparameter combinations on model performance. Additionally, a comparative analysis with prior studies on posture recognition using the MPII dataset showcases the superior efficacy of the proposed approach. In conclusion, the study provides an exhaustive analysis of the proposed three-phase model for decision support in human posture recognition. It underscores the indispensable role of transfer learning, image data augmentation, and

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hyperparameter optimization in achieving stellar accuracy rates in posture classification. With its comprehensive insights and innovative methodologies, the paper constitutes a seminal contribution to the field of human posture recognition and deep learning techniques [2].

Existing approaches to posture detection and correction vary widely in their methodologies and effectiveness. Traditional methods often rely on manual assessment by healthcare professionals or the use of wearable sensors to monitor posture in real-time. While these approaches can provide valuable insights, they often suffer from limitations such as high cost, inconvenience, and lack of scalability [12][17]. In recent years, advancements in computer vision and machine learning have paved the way for more automated and accessible solutions. Some existing approaches leverage deep learning techniques to analyze images or video streams in real-time, enabling posture assessment without the need for specialized hardware or manual intervention. These approaches typically involve training convolutional neural networks (CNNs) on large datasets of labeled images to recognize different postures accurately [11][15]. However, despite their promise, existing approaches still face several challenges. One common issue is the need for large and diverse datasets to train accurate models capable of generalizing across different individuals and environments [12][13][16][17]. Additionally, real-time performance can be a concern, particularly in applications requiring low latency, such as interactive feedback systems or assistive technologies. Moreover, many existing solutions focus primarily on posture assessment and fail to provide actionable feedback or intervention strategies to help users correct their posture effectively. This limitation underscores the importance of not only accurately detecting poor posture but also guiding users towards adopting healthier habits through feedback mechanisms. Overall, while existing approaches have made significant strides in posture detection and correction, there remains ample room for improvement in terms of accuracy, real-time performance, and user-centric design. Addressing these challenges could lead to more effective and accessible solutions for promoting better posture and overall musculoskeletal health.

III. METHODOLOGY

The methodology employed in this research project comprises several key stages, each designed to systematically address the objectives and challenges of the study. The following overview delineates the sequential steps undertaken:

A. Data Acquisition & Processing

In this phase of implementation, we acquired the required dataset for our posture classification model, labeled it for implementing a supervised learning algorithm, pre-processed it so as to train a model on this data. Novelty of our problem statement led to unavailability of reliable dataset for our use. Very limited work has been done in this domain and the dataset used for those models was dumped rather than making them public because of privacy issues. We had to create our very own dataset collection software for acquiring and labeling the required data for model training. The details of dataset are presented in Table 1. This collected dataset is analyzed, filtered and pre-processed into our final dataset. The software is used by 9 people of both the genders and wearing different apparels to make the dataset more diverse. The final dataset which we collected have 40K images classified into five classes namely, “Normal”, “Neck”, “Shoulders”, “Forward Bend”, “Backward Bend”.

The collected dataset is then pre-processed to fit training model onto it. Firstly, all the images are converted to grayscale and then are resized to a constant 256 X 256 px size. The resized images are then normalized so that their pixel values lie between the range 0-1 rather than lying in range 0-255.

TABLE 1: DETAILS OF DATASET

Class	Count of Images
Normal	7993
Neck	8015
Shoulders	8040
Forward Bend	8030
Backward Bend	8033
TOTAL	40111

B. Posture Detection Algorithm

This step involves the main algorithm which is responsible for detecting and classifying human postures as good/bad or posture specific classifications. Without the implementation of this algorithm, the system makes no sense, so this was the most crucial, challenging yet important step of our project implementation pipeline. In the context of our research, this posture detection algorithm utilizes computer vision techniques and machine learning models to analyze images captured by a webcam or camera feed. The algorithm’s aim is to accurately classify various postures-based features extracted from the images.

A Convolutional Neural Network was trained on the dataset to classify them into different postures. Several iterations and various model architectures were used to obtain a high level of accuracy and reliability. These iterations had a severe impact on choosing the right hyperparameters, data and classes for classification. We trained our CNN model using a large dataset of labeled posture images just collected from our own script. During the training process, the CNN learned to map input images to corresponding posture categories by adjusting the weights of its internal layers. CNN’s flexibility and scalability allowed us to tailor the model architecture and dataset size to suit the complexity of our posture detection task. Additionally, overfitting and avoiding techniques offered by CNN like Dropout enabled us to enhance model’s performance and robustness.

Our dataset which was collected and annotated for five classes as shown in the Table 1, was now changed and made to be a three-class dataset (as presented in Table 2). On training, it was observed that classes “Shoulder” and “Backward Bend” were contributing in reducing model’s accuracy and generalization capabilities. Due to the class “Backward Bend”, our model was classifying a relaxed position as abnormal position which should be counted as a good posture instead. Due to these reasons, these two classes were removed from our dataset.

TABLE 2: FINAL DATASET USED FOR TRAINING

Class	Count of Images
Normal	7924
Neck	5718
Forward Bend	7825
TOTAL	21467

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Our final dataset, used for training included a total of approx. 21K images classified into three classes. A CNN multi classification model was then trained on this final dataset. Further, the overall architecture of the model is presented in Figure 2.

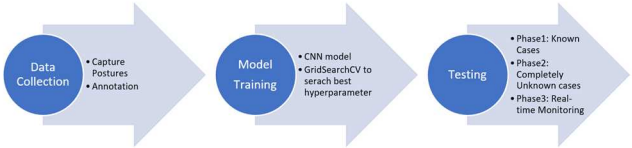

Fig 2: Overall Flow diagram

The algorithm for sitting posture detection is given in Algorithm 1 below:

Algorithm 1:

Input: Pre-recorded Video or video in Real-Time

Output: Posture

1. Collect data and annotation
2. Divide the dataset as Training set, validation set and test set.
3. Apply Grid search with CNN to train with best hyperparameters.
4. Test the trained model

C. *Model Architecture*

The model takes an image of shape as (256, 256, 1) as input. It has approx. 60 Million trainable parameters. Table 3 demonstrates the details of the architecture used for training the data.

TABLE 3: MODEL ARCHITECTURE

Layer Number	Layer Name	Size	Filters	Activation	Rate	Number of Parameters
1	Input Layer (Conv 2d)	(3, 3)	64	Relu	-	640
2	MaxPool2D	(2, 2)	-	-	-	0
3	Dropout	-	-	-	0.25	0
4	Conv2D	(3, 3)	128	Relu	-	73856
5	MaxPool2d	(2, 2)	-	-	-	0
6	Dropout	-	-	-	0.25	0
7	Conv2D	(3, 3)	256	Relu	-	295168
8	MaxPool2D	(2, 2)	-	-	-	0
9	Dropout	-	-	-	0.25	0
10	Flatten	-	-	-	-	0
11	Dense	256	-	Relu	-	58982656
12	Dropout	-	-	-	0.5	0
13	Dense	128	-	Relu	-	32896
14	Dropout	-	-	-	0.5	0
15	Output Layer (Dense)	3	-	Soft max	-	387

D. *Training*

The model was trained on the DGX Workstation which is a powerful computing facility provided by Nvidia that has 4 GPU's with several Cuda and Tensor cores per GPU along with a massive memory of around 128 GB. After uploading all the necessary files to the workstation, DGX was able to train and tune our model very efficiently.

DGX Workstation Specifications:

- Processor: 20-core intel Xeon e5-2698 v4 2.2 GHz
- System Memory: 256 GB DDR4
- GPU: 4 x GPUs (Tesla V100 SXM2)
- GPU Memory: 128 GB (8X32) total system Cuda cores: 5000/GPU Tensor cores: 600/GPU
- Storage: 4X 1.92 TB SSD RAID 0
- OS: Ubuntu Linux

Now, the dataset was divided into training, testing and validation sets for training and evaluation.

- Training Set: 12800 Images
- Testing Set: 4000 Images
- Validation Set: 4667 Images

The best parameters as received after employing hyper-paramter tuning techniques like RandomizedSearchCV and GridSearchCV were as follows:

- Batch Size: 128
- Number of Epochs: 10
- Optimizer: "adam"
- Loss: "Categorical_CrossEntopy"

Figure 3 presents the Loss Vs Epoch graph for training and validation.

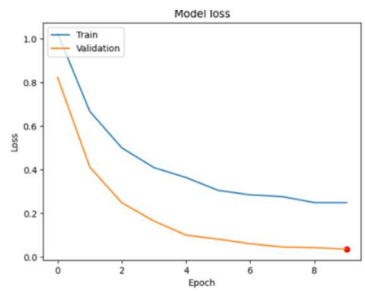

Fig 3: Model's Loss v/s Number of Epochs

E. *Performance Evaluation & Inference*

The testing and evaluation for this research was divided into three phases. Each phase's motive was to test the robustness of the model.

Phase 1: In the first phase of testing, the model was tested on the test dataset we split earlier. In this method the test dataset was loaded and pre-processed so as to fit our model required shape and output by the model were compared with the ground truth values which were the original labels of each image. This was done for our testing set of 4000 images and results were printed using a classification report (Table 4). The classification report gave us insights about the model's precision, recall, and F1 score.

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TABLE 4: TESTING PHASE 1

Type	Precision	Recall	F1-Score
Class Normal	0.96	1.00	0.98
Class Neck	1.00	0.97	0.98
Class Forward Bend	1.00	0.99	1.00
Micro Average	0.99	0.99	0.99
Macro Average	0.99	0.99	0.99
Weighted Average	0.99	0.99	0.99
Sample Average	0.99	0.99	0.99

Phase 2: In the second phase of testing the model, a new dataset was collected from new volunteers who were not a part of the training data collection process. Through this method, we were able to identify how well our model was performing on truly unseen data and that too collected in a newer environment. This was done for a collected dataset of around 7000 images and results were printed using a classification report (Table 5). The classification report gave us insights about the model's precision, recall, and F1 score.

TABLE 5: TESTING PHASE 2

Type	Precision	Recall	F1-Score
Class Normal	0.72	0.50	0.59
Class Neck	0.61	0.57	0.59
Class Forward Bend	0.60	0.78	0.68
Micro Average	0.63	0.63	0.63
Macro Average	0.64	0.62	0.62
Weighted Average	0.64	0.63	0.62
Sample Average	0.63	0.63	0.63

Phase 3: In the third phase of the inference and evaluation process, a python script was written to open the user's webcam and capture frames every 10 milliseconds. These captured frames were then processed to fit our model's requirement and then the predictions were displayed on the user's screen (as shown in Figure 4 and Figure 5) so that the user can manually assess the working and accuracy of the model. Further, Table 6 presents the detailed results.

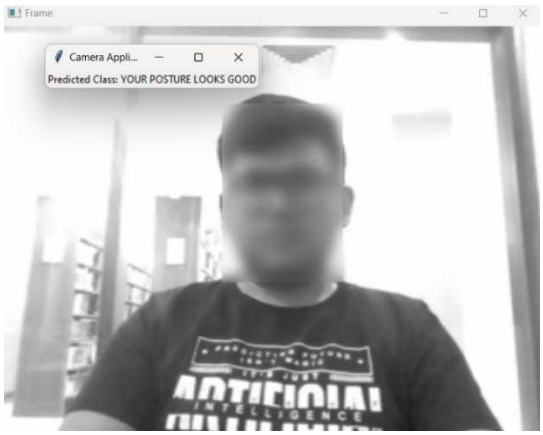

Fig 4: Testing Phase 3

TABLE 6: TESTING PHASE 3

User	Total Frames	Normal (Correct)	Neck (Correct)	Forward (Correct)	Normal (Incorrect)	Neck (Incorrect)	Forward (Incorrect)	Accuracy (%)
1	120	38	25	24	2	15	16	72.5
2	120	36	27	19	4	13	21	68.3
3	120	37	24	26	3	16	14	72.5
4	120	39	31	24	1	9	16	78.3
5	120	34	32	21	6	8	19	72.5
6	120	37	28	20	3	12	20	70.8
7	120	33	29	23	7	11	17	70.8

IV. RESULTS

The approach followed in this research has led to the development of a model for real time human posture classification which uses simple hardware, the webcam already embedded in the user's device and avoid using specialized hardware like pressure sensors and smart chairs. Through this approach, each class was getting predicted with an average accuracy of more than 60% which is understandable, given the smaller and less diverse dataset collected. This research proves the possibility of solving this major problem of posture detection without the use of special devices. Majority of the posture detection algorithms, rely on external hardware to detect human postures in real time which makes them really hard to scale. When, we are solving a problem like musculoskeletal disorders, which is of greatest importance in today's digital world, scalability is the main factor to take care of while devising a solution. Lack of such external hardware makes this algorithm scalable and easily adaptable by the end users. The algorithm devised in our research was also deployed for real life use using Tkinter GUI windows application which was deployed as a windows installer using PyInstaller and INNO Setup.

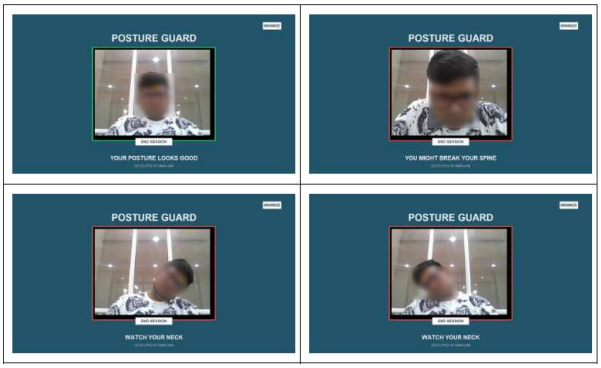

Fig 5: Posture Guard Application

V. CONCLUSION

In conclusion, this research paper presents a groundbreaking model for decision making in front sitting human posture recognition. Through rigorous experimentation and meticulous methodology, the paper navigates the challenges posed by deep CNN models and provides innovative solutions to enhance classification performance. This study lays the groundwork for future advancements in this domain, eliminating the necessity for specialized hardware such as smart chairs or pressure sensors.

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Utilizing potent deep learning techniques like CNN, the posture detection challenge can be effectively addressed, contingent upon the availability of a comprehensive and diverse dataset. Subsequent research endeavors could focus on expanding the scope of data collection efforts to encompass a broader demographic spectrum, including individuals of varying genders, races, ethnicities, body types, and attire. This diversification aims to enhance the model's adaptability to unseen data instances. While our model demonstrated a precision of approximately 0.99 on test data from contributors to the dataset, precision dropped to around 0.60 when applied to new individuals, underscoring the necessity for improved and more expansive data collection methods.

REFERENCES

[1] Peng, Y., Peng, J., Li, J., Yao, C., & Shi, X. (2019). Smart home based on kinect gesture recognition technology. International Journal of Performability Engineering, 15(1), 261.

[2] R. O. Ogundokun, R. Maskeliūnas, and R. Damaševičius. 2022. Human Posture Detection Using Image Augmentation and Hyperparameter-Optimized Transfer Learning Algorithms. Applied Sciences 12, 19 (2022), 10156. https://doi.org/10.3390/app121910156.

[3] N. Mishra, I. Mayatra, K. Patel, and A. Mishra. 2020. A study to find work related musculoskeletal disorders and associated risk factors among dairy workers. 5, 21–24.

[4] M. Martínez-Estrada, T. Vuohijoki, A. Poberznik, A. Shaikh, J. Virkki, I. Gil, and R. Fernández-García. 2023. A Smart Chair to Monitor Sitting Posture by Capacitive Textile Sensors. Materials 16, 13 (2023), 4838. https://doi.org/10.3390/ma16134838.

[5] X. Huang, Y. Xue, S. Ren, and F. Wang. 2023. Sensor-Based Wearable Systems for Monitoring Human Motion and Posture: A Review. Sensors 23, 22 (2023), 9047. https://doi.org/10.3390/s23229047.

[6] X. Li, P. Zhao, M. Wu, Z. Chen, and L. Zhang. 2021. Deep learning for human activity recognition. Neurocomputing 444 (2021), 214–216. https://doi.org/10.1016/j.neucom.2020.11.020.

[7] A. Gomez-Marin, N. Partoune, G. J. Stephens, and M. Louis. 2012. Automated Tracking of Animal Posture and Movement during Exploration and Sensory Orientation Behaviors. PLOS ONE 7, 8 (2012), e41642. https://doi.org/10.1371/journal.pone.0041642.

[8] E. Piñero-Fuentes, S. Canas-Moreno, A. Rios-Navarro, M. Domínguez-Morales, J. L. Sevillano, and A. Linares-Barranco. 2021. A Deep-Learning Based Posture Detection System for Preventing Telework-Related Musculoskeletal Disorders. Sensors 21, 15 (2021), 5236. https://doi.org/10.3390/s21155236.

[9] A. Kulikajevas, R. Maskeliunas, and R. Damaševičius. 2021. Detection of sitting posture using hierarchical image composition and deep learning. PeerJComput Sci. 7 (2021), e442. https://doi.org/10.7717/peerj-cs.442.

[10] Y. Byeon et al. 2020. Posture Recognition Using Ensemble Deep Models under Various Home Environments. Applied Sciences.

[11] O. Roseline et al. 2020. Human Posture Detection Using Image Augmentation and Hyperparameter-Optimized Transfer Learning Algorithms. MDPI Journals.

[12] A. Audrius et al. 2021. Detection of Sitting Posture Using Hierarchical Image Composition and Deep Learning. PeerJ Computer Science Journals.

[13] E. Enrique et al. 2021. A Deep-Learning Based Posture Detection System for Preventing Telework-Related Musculoskeletal Disorders. MDPI Journals.

[14] M. Devaraj et al. 2023. Modelling Proper and Improper Sitting Posture of Computer Users Using Machine Vision for a Human–Computer Intelligent Interactive System during COVID-19. Applied Sciences.

[15] J. Estrada et al. 2023. Computer Users Sitting Posture Classification Using Distinct Feature Points and Small Scale Convolutional Neural Network for Humana Computer Intelligent Interactive System During COVID-19. Journal of Computer Science.

[16] M. Schmid et al. 2023. Recurrent Network Solutions for Human Posture Recognition Based on Kinect Skeletal Data. Sensors.

[17] M. Isaac et al. 2024. A Method for Complex Posture Recognition During Long-Term Sitting Using Neural Networks and Pressure Mapping Systems. Science Direct.

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